

A BRIEFING FOR STATISTICIANS, IT TEAMS AND MANAGEMENT

AI Infrastructure for Statistical Offices

What it really takes to run AI in a national statistical office — with a focus on the African context

Compute

Data & MLOps

Deployment

Governance

Cost

People & roadmap

Data Innovation For Africa (DI4A) - African Development Bank

Why this briefing, and what you will learn

Objective: give a shared, practical basis for deciding how to build AI capability in an NSO

THE CORE MESSAGE

An AI capability in a statistical office is a production platform with legal obligations — not a model and not a one-off project.

WHO IT IS FOR



Statisticians and methodologists



IT, platform and security teams



Management and oversight bodies

BY THE END YOU WILL BE ABLE TO

1

Explain the five layers of an AI system and why the model is the smallest part

2

Match hardware, hosting and models to each workload and its data classification

3

Identify the new confidentiality risks AI creates and the controls that stop them

4

Estimate the full five-year cost, knowing which decisions really move the total

5

Plan the team, the maturity steps and a gated 24-month roadmap

Agenda

01



Framing & architecture

Why the model is the easy part, use cases, the African context and the five-layer stack

02



Compute, data & MLOps

AI hardware and sizing, power, data platforms, labelled data, retrieval, reproducibility

03



Deployment choices

Four hosting archetypes, a decision framework, build vs. buy, open vs. proprietary

04



Governance & confidentiality

New disclosure risks, the legal landscape, classification tiers and controls

05



Cost & budgeting

The full cost stack, five-year scenarios, break-even points and the budget case

06



People & roadmap

Roles, skills, maturity levels, a 24-month plan and a readiness checklist

Ten key concepts in plain language

Keep this page in mind — every later slide uses these words



Model

A program whose behaviour is learned from examples rather than written as rules.



Training

Showing a model many labelled examples so it learns. Heavy, occasional compute.



Inference

Using a trained model to produce an answer. Light per request, but continuous.



GPU

A chip doing many calculations in parallel. Vital for large models, optional for classical ML.



LLM

Large language model: a generative model trained on vast text that drafts, extracts and classifies.



Open-weight model

A model whose parameters you can download and run on your own servers.



Fine-tuning

Extra training of an existing model on your own data to specialise it.



RAG

Retrieval-augmented generation: the model answers using passages found in your documents.



MLOps

The practices and tools that take a model from experiment to monitored production.



Gold standard

A frozen, expertly coded dataset used to measure how well a model performs.

SECTION 01

01

Framing & architecture

Why is the model the easiest part of an AI system?

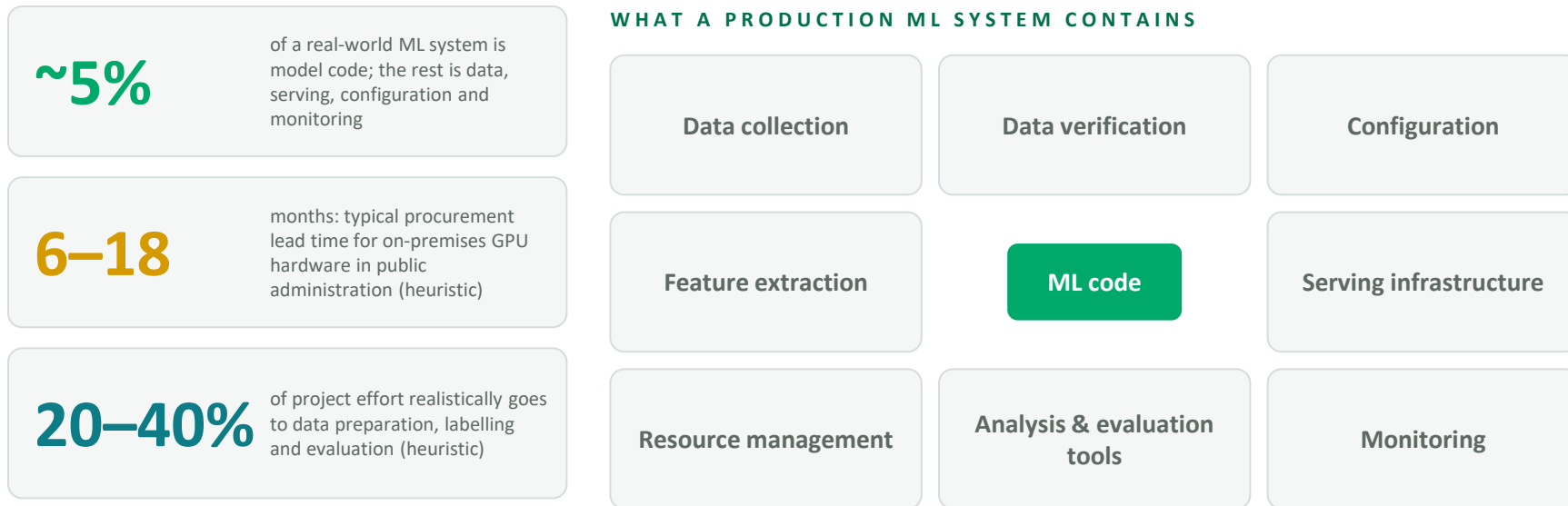
IN THIS SECTION

- › The model is the easy part
- › Where AI earns its place
- › A worked example
- › NSO vs. technology company
- › The African context
- › The five-layer stack



The model is the easy part

In a production AI system, the modelling code is a small box surrounded by much larger ones



KEY POINT An AI capability is a platform, not a project: it needs recurrent funding, an owner and an operations team — and success is measured against the current manual process, not against perfection.

Source: Sculley et al., "Hidden Technical Debt in Machine Learning Systems", NeurIPS 2015 (diagram adapted). Lead-time and effort ranges are planning heuristics.

Where AI earns its place in official statistics

Eight families of use case, ordered roughly by maturity across statistical offices



Automatic coding

Free-text answers coded to ISCO, ISIC, COICOP, CPC or ICD classifications.

Established



Web & scanner data

Online prices for the CPI, product matching to COICOP, online job adverts.

Established



Record linkage

Matching census, business registers and administrative files; deduplication.

Established



Editing & imputation

Selective editing, outlier detection and model-based imputation.

Established



Text & documents

Survey open-ends, register enrichment and metadata generation.

Emerging



Imagery & geospatial

Building footprints for census mapping, land cover, crop area estimation.

Emerging



Assistive tooling

Code assistance, drafting, translation and search over methodology.

Emerging



Confidentiality support

Disclosure-risk assessment and synthetic data generation.

Needs sign-off

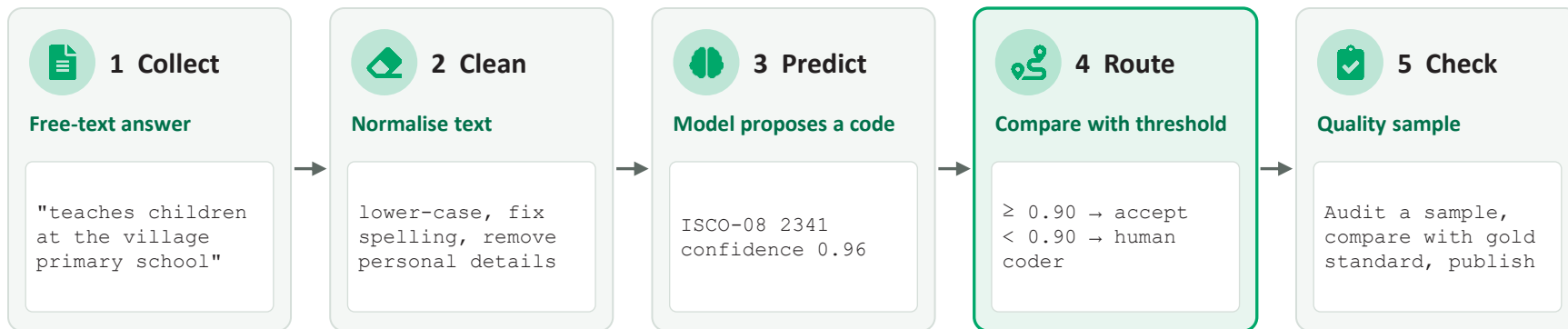


KEY POINT

Start where you already hold years of manually coded answers — usually coding and classification. It can be evaluated honestly on day one.

Worked example: coding occupations with AI

Follow one census answer from questionnaire to published table



ILLUSTRATIVE: 100,000 CENSUS ANSWERS

70,000 coded automatically

30,000 to coders

Coders focus on the hard 30%, plus quality checks on the rest.

THREE LESSONS

- ✓ The confidence score matters as much as the code
- ✓ Set the threshold on a frozen gold standard
- ✓ People are redeployed, not removed

Illustrative teaching figures. Real automation shares depend on data quality, the classification and the chosen threshold. ISCO-08 2341 = primary school teachers.

A statistical office is not a technology company

The differences are legal and methodological — not cultural

DIMENSION	TYPICAL TECHNOLOGY COMPANY	NATIONAL STATISTICAL OFFICE
 Data	Mostly first-party, consented, replaceable	Legally mandated collection; confidential microdata under statistical secrecy
 Tolerance for error	Ship, measure, iterate; roll back quietly	A published figure is a public act; errors trigger revisions and damage trust
 Release cadence	Continuous deployment, weekly	Release calendar published in advance; formal revisions policy
 Reproducibility	Desirable	Mandatory — a figure may have to be reproduced years later, on demand
 Explainability	Only where a regulator asks	Methodological documentation, peer review, international comparability
 Talent market	Competes on salary	Public pay scales; retention through mission, data access and training
 Infrastructure	Elastic public cloud by default	Sovereignty, procurement law, classification tiers, unreliable power in places

What changes for African statistical offices

Four facts that shape every infrastructure decision on the continent

~600 M 

people in sub-Saharan Africa without electricity in 2024 (47% of the population)

IEA

0.6% 

Africa's share of global data-centre capacity; power is now the main constraint

ADCA 2026

40+ 

African countries have enacted data protection laws

ADCA 2026

< 2% 

of Africa's ~2,000 languages are adequately supported by large language models

BCG 2026

WHAT IT MEANS IN PRACTICE



Power first

Check kW capacity and backup autonomy before choosing hardware.



Hosting is a sovereignty choice

Few in-country options; cross-border hosting needs a legal basis.



An uneven legal map

Align with your statistics act, national data law and AU frameworks.



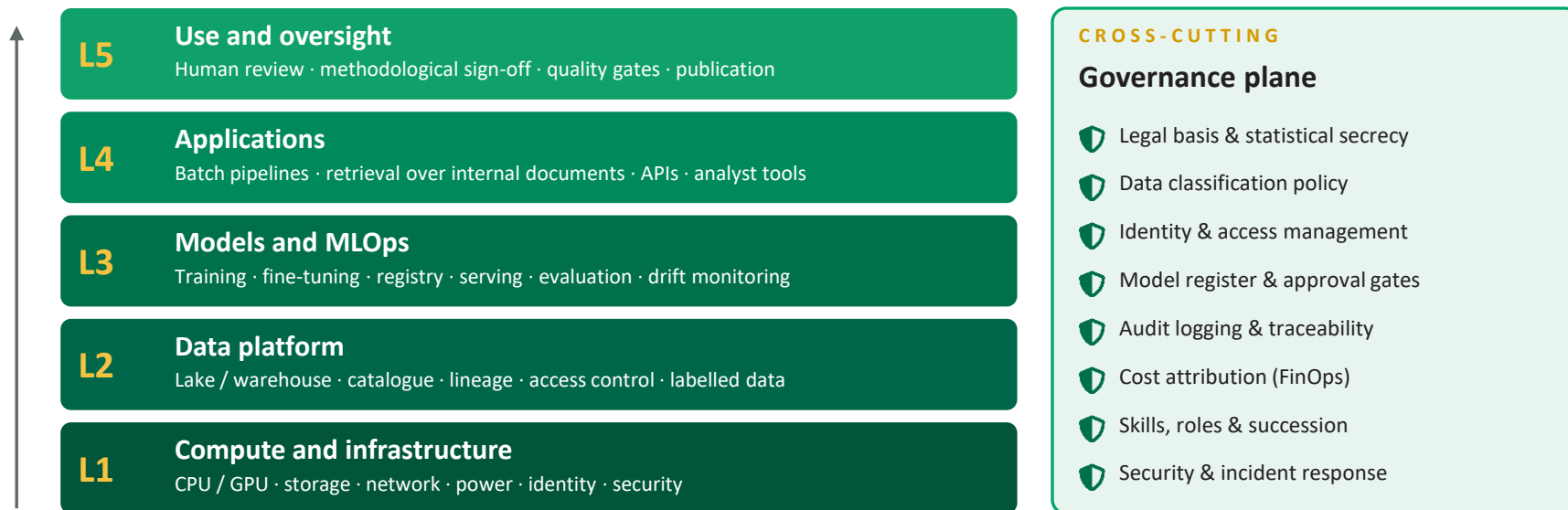
The opportunity

Most value comes from light inference and classical ML — no giant clusters needed.

Sources: IEA (2025); Africa Data Centres Association (ADCA), Data Centres in Africa 2026; BCG, Advancing Africa's AI and Digital Economy (2026).

Five layers, one governance plane

Read it bottom-up: each layer is only as useful as the one beneath it



KEY POINT

The governance plane is not added at the end. It constrains every layer, and it is where most unbudgeted effort hides.

SECTION 02

02

Compute, data & MLOps

What hardware, data and processes must exist before a model can be trusted?

IN THIS SECTION

- › Training vs. inference
- › Matching hardware to workload
- › AI hardware explained
- › Power, storage, network
- › Designing for unreliable power
- › The data platform
- › Labelled data
- › How retrieval (RAG) works
- › MLOps and environments



Training vs. inference: two very different loads

Most hardware mistakes come from confusing the two



Training

Like teaching a new coder the classification manual

How often	A few times a year
Compute	Heavy: GPUs for hours to days
Data needed	Thousands of labelled examples
Main risk	Buying a cluster that sits idle
Where to run	Rent or borrow burst capacity



Inference

Like the trained coder processing forms every day

How often	Continuously, every day
Compute	Light per request: CPU or one GPU
Data needed	The new records to classify
Main risk	Downtime during a release window
Where to run	Stable, owned or contracted capacity



KEY POINT Most NSO use cases are inference-heavy and training-light. Size owned hardware for steady inference; rent training capacity when you need it.

Match the hardware to the workload

The most expensive mistake: buying training-class hardware for an inference-class problem

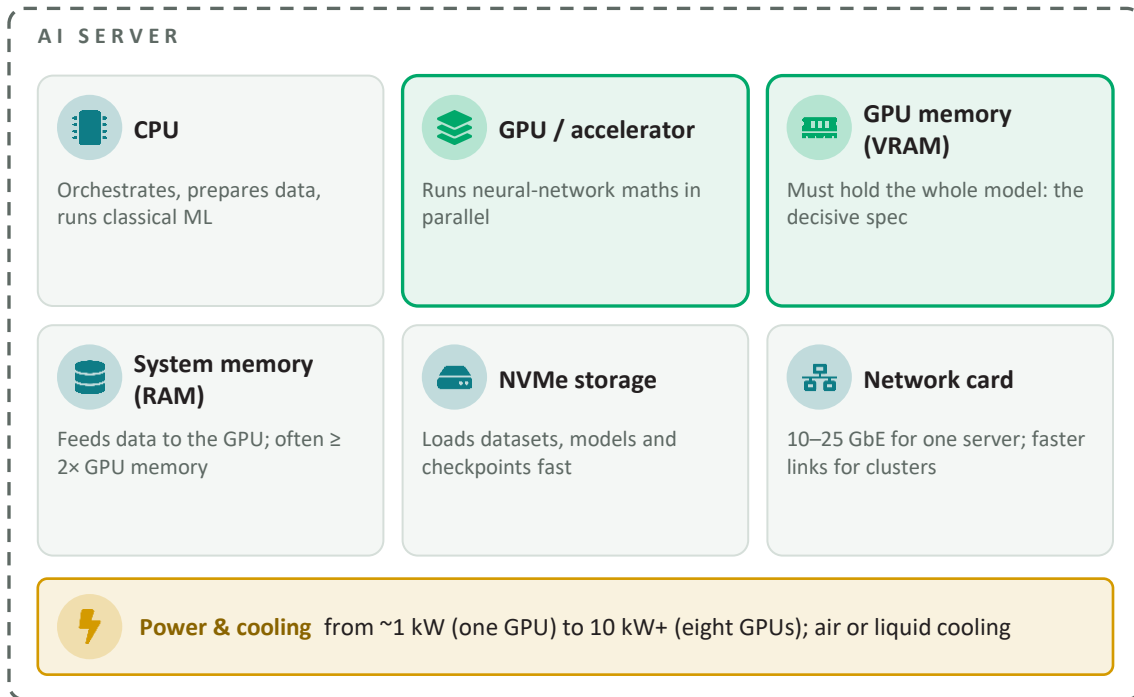
WORKLOAD	TYPICAL HARDWARE	DURATION	FOR AN NSO?	COMMENT
Classical ML (gradient boosting, linear models)	CPU only, 16–64 cores	Minutes–hours	Yes	Covers much of coding, editing, linkage
Fine-tuning small text models (under 1B parameters)	1 GPU, 24–48 GB	Hours	Yes	The workhorse for text classification
Running small open LLMs (7–14B, compressed)	1 GPU, 24–48 GB	Real time, teams	Yes	Internal assistance, retrieval, extraction
Running larger open LLMs (~70B)	2–4 GPUs, 80 GB each	Real time, office	Sometimes	Compare honestly with a managed API
Efficient fine-tuning of ~70B models (LoRA)	4–8 GPUs, 80 GB each	Days	Rarely	Only with a strong domain case
Pre-training a foundation model	Hundreds–thousands of GPUs	Weeks–months	No	Not an activity for a statistical office



KEY POINT Before requesting a GPU, answer three questions: is it training or inference? What is the sustained load versus the peak? Does the data classification even permit external compute?

What is inside an AI server?

Seven components — and why each one matters for AI workloads



AN ANALOGY



CPU = a few expert chefs doing complex tasks one by one



GPU = thousands of line cooks each doing one simple task at once

WHY IT MATTERS

Neural networks repeat one simple calculation billions of times — ideal for GPUs. Classical ML stays on CPUs.

GPU memory is the spec that decides

A simple calculation tells you which models fit on which card

THE RULE OF THUMB

Memory $\approx$ parameters $\times$ bytes per parameter + ~20%

BYTES PER PARAMETER

● 16-bit (full) **2**

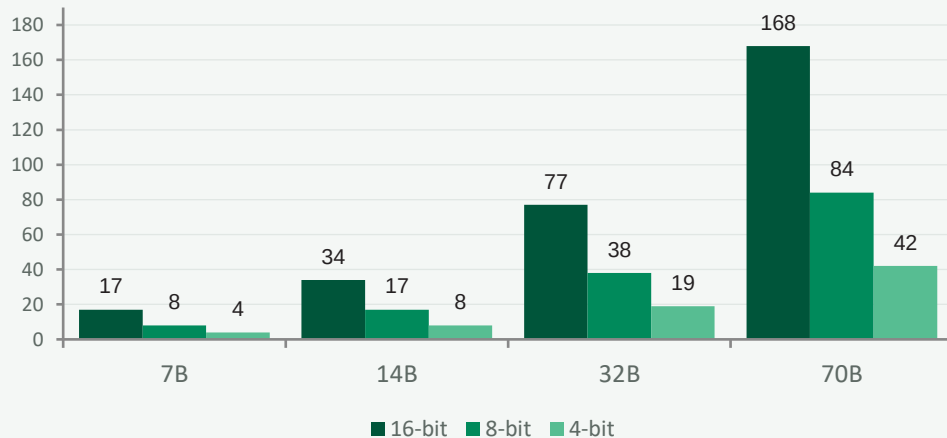
● 8-bit (compressed) **1**

● 4-bit (compressed) **0.5**



Training needs far more: full fine-tuning takes ~16–20 bytes per parameter. LoRA / QLoRA make it fit on one card.

Memory needed to run a model (GB, incl. ~20% overhead)



COMMON CARD SIZES

24 GB

entry

48 GB

inference

80 GB

H100

96 GB

RTX PRO 6000

141 GB

H200

180 GB

B200

Accelerator families at a glance

Examples are illustrative, not endorsements — product lines change every 12–18 months

CLASS	EXAMPLES (2026)	MEMORY	POWER	FOR AN NSO	TYPICAL USE
 CPU servers	Current x86 or Arm server processors	System RAM	0.3–1 kW	Core	Classical ML, small-model inference
 Entry / inference GPUs	NVIDIA L4, L40S	24–48 GB	~70–350 W	Good fit	Text classifiers, 7–14B LLMs
 Workstation-class GPUs	NVIDIA RTX PRO 6000 (Blackwell)	96 GB	~600 W	Good fit	Up to 70B compressed; LoRA
 Data-centre GPUs	NVIDIA H100 / H200 / B200; AMD Instinct	80 / 141 / 180 GB	~0.7–1 kW each	Selective	Large models, multi-GPU training
 Desk-side AI systems	NVIDIA DGX Spark	128 GB shared	Desk power	Good fit	Prototyping, small teams
 Cloud-only accelerators	Google TPU, AWS Trainium / Inferentia	Rented	Provider's	If cloud allowed	Only for permitted data tiers



KEY POINT

Check software support before brand: the tools your team already uses must run on the card. Memory first, speed second.

Sources: NVIDIA product pages and vendor datasheets as compiled by GetDeploying (2026). Power figures are indicative maximums.

Three reference configurations for an NSO

Start small, grow only when a use case passes its gate — and rent the top tier rather than buy it

STARTER

LOOKS LIKE

1 server, 1–2 GPUs of 24–48 GB, or a desk-side AI system

RUNS

Classical ML, text classifiers, 7–14B LLMs for a team

POWER & COOLING

~1–2 kW; standard air cooling

STAFF

Existing IT + 0.5 platform engineer

HARDWARE COST

Tens of thousands of USD

STANDARD

LOOKS LIKE

1–2 servers, 2–4 GPUs of 48–96 GB each

RUNS

Office-wide 30–70B models, LoRA fine-tuning, several production uses

POWER & COOLING

~3–5 kW per server; dedicated rack

STAFF

1–2 platform engineers

HARDWARE COST

Roughly USD 50–150k

ADVANCED

LOOKS LIKE

8-GPU data-centre nodes (80–180 GB each), fast links

RUNS

Large-model fine-tuning, many teams, research

POWER & COOLING

6–10 kW+ per node; often liquid cooling

STAFF

2+ engineers, 24/7 facilities support

HARDWARE COST

USD 300k+ per node



KEY POINT

Most offices should start at Starter, move to Standard after a use case passes its gate, and rent or share Advanced capacity.

Indicative 2026 hardware prices only, excluding facility works, software, staff, import duties and currency effects. Obtain current quotes.

Buying AI hardware: a procurement checklist

What to write into the tender — and the traps to avoid



SPECIFY IN THE TENDER

- ✓ GPU memory per card and in total, sized to your target models
- ✓ Support for the software your team uses (e.g. CUDA, ROCm)
- ✓ Power draw per server and cooling type (air or liquid)
- ✓ In-country warranty, on-site support and spare-part lead times
- ✓ Error-correcting (ECC) memory and remote management
- ✓ Security firmware updates for at least five years
- ✓ Confirmed export licensing and delivery conditions



AVOID

- ✗ Buying the newest flagship "to be future-proof"
- ✗ Consumer gaming cards in production servers
- ✗ Grey-market or unwarranted refurbished units
- ✗ Specifying hardware before the power audit
- ✗ Proprietary software that locks you to one vendor
- ✗ No budget for refresh, disposal and e-waste



GOLDEN RULE

Rent before you buy: run the pilot on rented or shared GPUs for 3–6 months, measure real utilisation, then decide.

The quiet bottlenecks

Rarely in the business case — and routinely the cause of months of delay



Storage tiering

01

Fast NVMe for active data, cheaper object storage for archives. Plan for 3–5× the raw data size in checkpoints and intermediate files.



Network fabric

02

10 GbE is the practical floor; 25–100 GbE for multi-node training. A slow internal network turns an expensive GPU into an idle one.



Power and cooling

03

A dense 8-GPU node can draw roughly 6–10 kW, while many server rooms are built for 3–5 kW per rack. This is a facilities budget, not IT.



Backup and recovery

04

Training snapshots, model weights and the model registry need tested backups. They are often outside existing disaster-recovery scope.



Data egress

05

In the cloud, data coming in is usually free and data going out is not. Moving microdata back and forth becomes a recurring, uncapped cost.



Identity integration

06

GPU servers and model gateways must use the corporate directory from day one. Retrofitting single sign-on later costs far more.

Designing for unreliable power and connectivity

Practical patterns that keep AI workloads running where the grid and the internet are not guaranteed



Know your power envelope

Measure kW per rack and backup autonomy (hours) before specifying any hardware.



Protect every run

UPS for clean shutdown, generator or solar-hybrid for autonomy; checkpoint long jobs.



Prefer efficient models

Small or compressed models on CPU or one GPU deliver most value at a fraction of the power.



Work offline-first

Local mirrors for packages, containers and model weights; no build depends on an international link.



Schedule batch work

Run heavy processing overnight or in stable-power windows; queue jobs rather than let them fail.



Share capacity

Government data centres, university computing centres or regional peers can host what one office cannot.



CONTEXT Mission 300 — a World Bank Group and African Development Bank initiative — aims to connect 300 million people in sub-Saharan Africa to electricity by 2030. Plan for today's grid, not tomorrow's.

Source: World Bank Group, Mission 300 (2024). Power figures for GPU nodes are indicative; check vendor specifications.

You cannot do AI on data you cannot find, trust or join

Six capabilities that must exist before a model is worth training



One governed data platform

Survey, administrative and alternative data under one access model — not hundreds of extracts on shared drives.



A metadata catalogue

Searchable business and technical metadata, aligned to GSIM and SDMX. If analysts cannot find a variable, they recreate it.



End-to-end lineage

Which input version, processed by which code version, produced which output.



Fine-grained access control

Row- and column-level rules mapped to classification tiers, enforced by the platform.



Versioned snapshots

Immutable, dated copies of the exact data used to train and test every model.



Labelled gold-standard sets

Curated training and test datasets with documented coding rules and a named owner.



KEY POINT If you can fund only one thing this year, fund the data platform. It is the only layer whose value does not depend on any AI technology surviving.

The binding constraint is labelled data, not GPUs

Compute can be rented in an afternoon. A defensible gold standard takes months

WHAT YOU ALREADY OWN

Years of manually coded responses

- ✓ Occupation and activity codes from past censuses and surveys
- ✓ Editing decisions recorded in production systems
- ✓ Past record-linkage decisions, including clerical review
- ✓ Coding manuals and quality reports explaining the labels

This archive is your single largest AI asset. Locate it, describe it and protect it first.

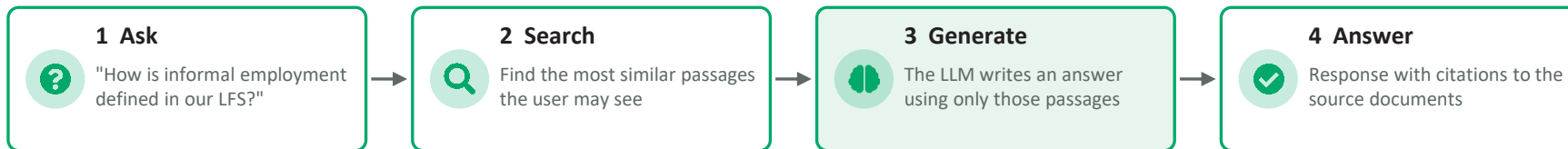
WHAT YOU STILL HAVE TO BUILD

- 1 Freeze the test set first**
Before any modelling, or every accuracy claim is contaminated.
- 2 Budget labelling as real work**
Coder time, guidelines, resolving disagreements, measured agreement.
- 3 Measure the human baseline**
Coders are not perfect either; their accuracy is the bar to clear.
- 4 Stratify for rare categories**
Overall accuracy hides failure in small, policy-relevant groups.
- 5 Maintain across revisions**
A gold standard on an outdated classification silently expires.

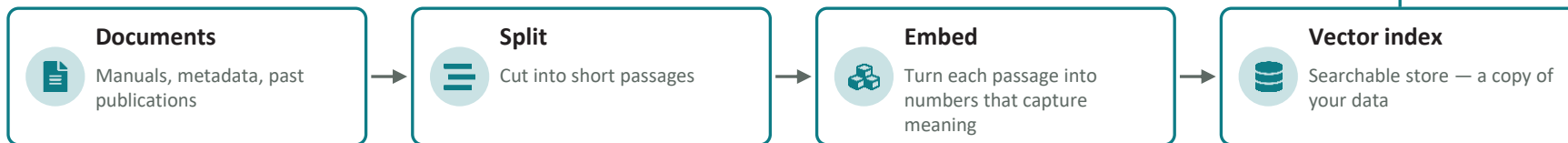
How retrieval-augmented generation (RAG) works

The most common way to make a language model answer from your own documents

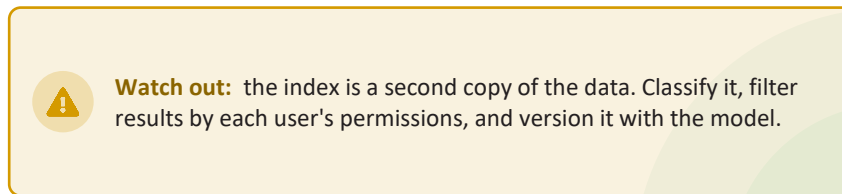
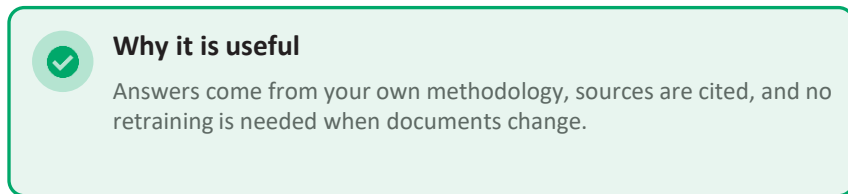
ANSWERING A QUESTION



PREPARED IN ADVANCE

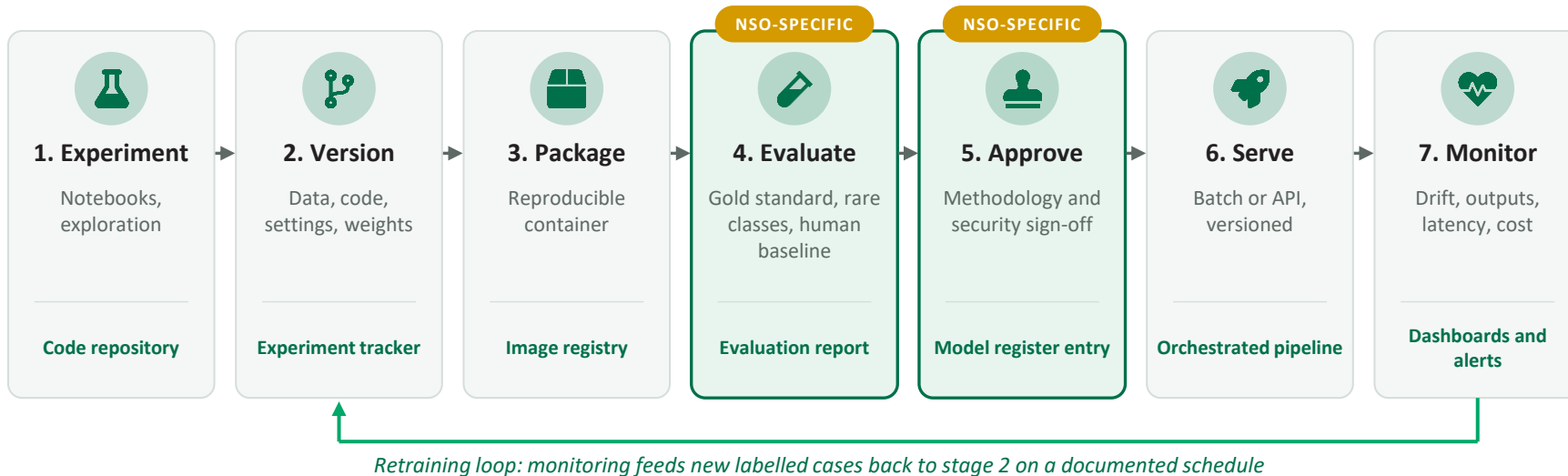


search uses the index



From notebook to production in seven stages

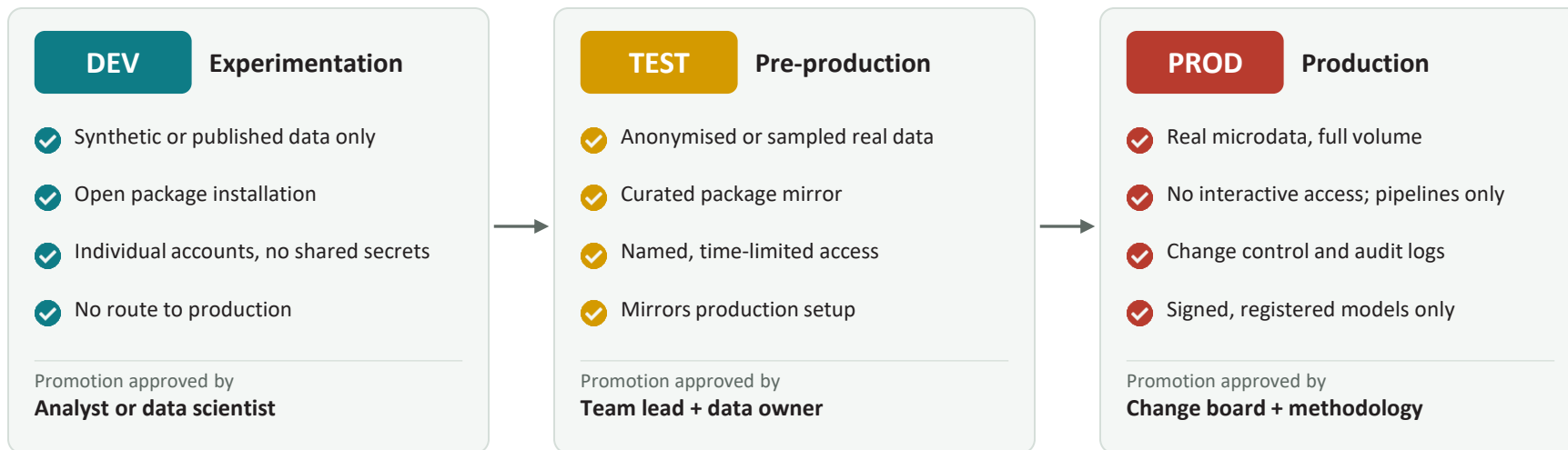
Each stage has an owner and an artefact. A pilot that skips stages 2–5 cannot be promoted



KEY POINT Stages 4 and 5 cannot be borrowed from industry practice. That is where methodology, not engineering, holds the pen.

Environments and the promotion path

Which data may be seen where — this one picture settles most analyst–security disputes



Plus one: the secure research zone

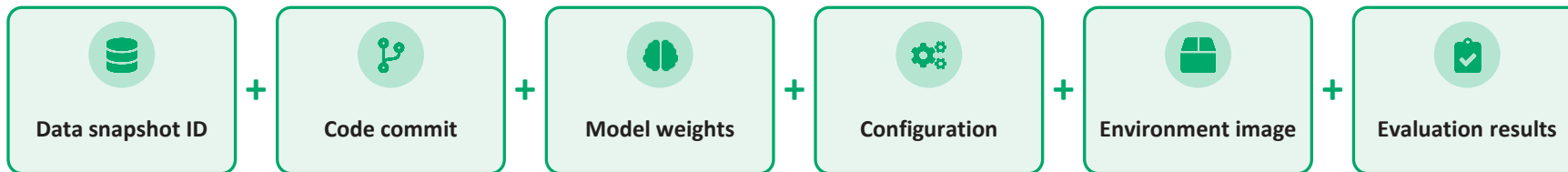


A separate enclave for work on identifiable microdata: no outbound network, outputs checked before release, all activity logged. Treat it as a fourth environment with its own model allow-list — never as an extension of DEV.

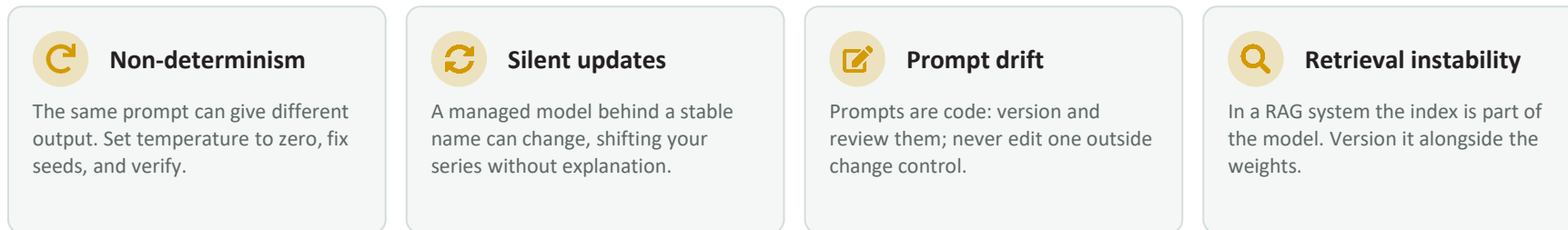
Reproducibility is a statistical obligation

You may be asked to reproduce a published figure years later — generative models make that harder

SIX THINGS TO PIN TOGETHER FOR EVERY PUBLISHED RUN



THE SPECIFIC HAZARDS OF GENERATIVE MODELS



 **KEY POINT** Where a figure must be reproducible on demand, use a self-hosted open-weight model you control — or keep generative AI off the critical path.

SECTION 03

03

Deployment choices

Where should each workload run — and who decides?

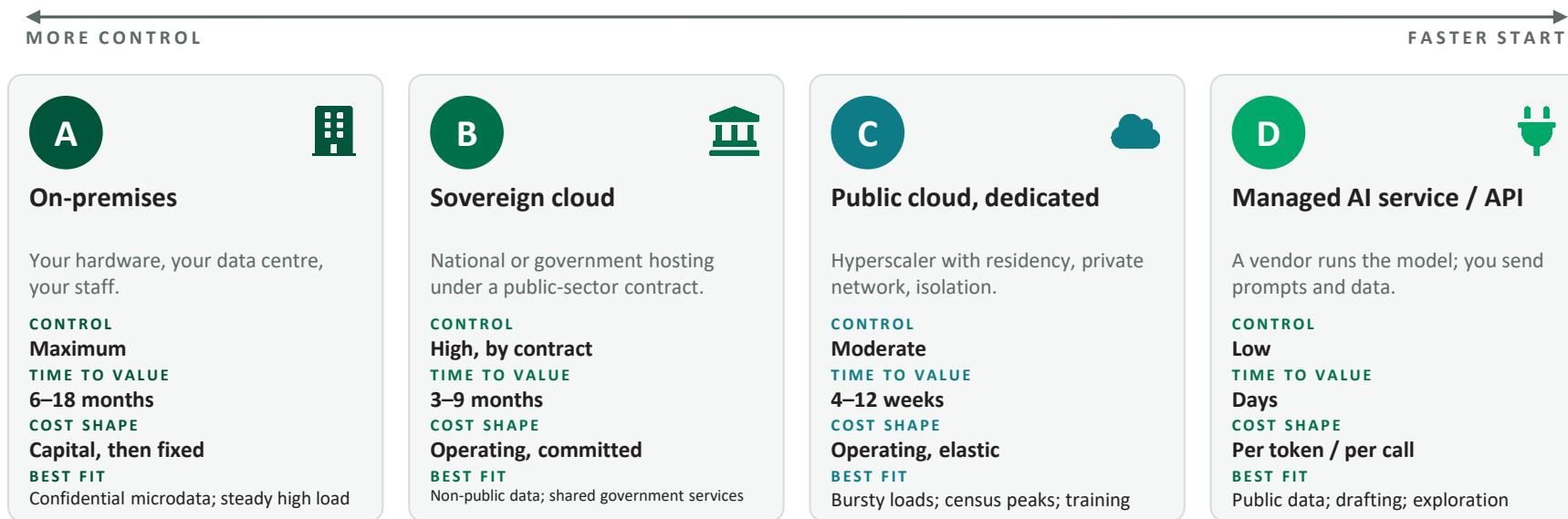
IN THIS SECTION

- › Four hosting archetypes
- › A defensible decision framework
- › On-premises: the full bill
- › Public cloud traps
- › Hosting options in Africa
- › The hybrid pattern
- › Build or buy
- › Open vs. proprietary models
- › Test yourself



Four deployment archetypes

Ordered from most control to fastest start. Most offices run several at once



KEY POINT The question is never "which one?" but "which workload belongs where?" — and data classification decides that before cost does.

A decision framework you can defend

Score seven drivers honestly. When they conflict, classification and law outrank cost

DRIVER	POINTS TO ON-PREMISES / SOVEREIGN	POINTS TO CLOUD / MANAGED	WEIGHT
 Data classification	Identifiable microdata under statistical secrecy	Published or already-public data	Decisive
 Legal basis & jurisdiction	Law restricts where or by whom data is processed	No restriction; contract and residency suffice	Decisive
 Load profile	Steady, predictable, high utilisation	Spiky or seasonal — census, training runs	High
 Time to first value	You can wait for procurement and build	You need a working pilot this quarter	High
 Budget mechanics	Capital (or donor) funding, little recurrent budget	Stable operating budget year to year	High
 In-house platform skills	You have, or can hire and keep, platform engineers	You do not, and cannot realistically retain them	High
 Exit cost & lock-in	Long-term capability you intend to own	Reversible commitment with open formats	Medium
 Read the two decisive rows first. If classification or law rules an option out, the other five drivers are irrelevant for that workload. Document the reasoning once and reuse it.			

What "we will just buy a server" really means

The hardware invoice is typically under half of the five-year cost of owning the capability

< 50%

of five-year cost is the hardware itself (planning heuristic)

The rest is facility, software, operations, refresh — and people.



Hardware

GPU nodes, fast storage, network, racks. Size for your workload, not the biggest model.



Facility

Power, UPS, generator, cooling, fire safety. Often a separate budget and approval chain.



Platform software

Scheduler, container registry, storage, identity, monitoring, secrets, backup.



Operations

Drivers, patching, capacity planning, on-call. Continuous work, not a one-off install.



Lifecycle

Refresh every 3–5 years. Fund the replacement at purchase, not in year four.



People

About 1.5–2 platform engineers for a small GPU estate, plus security and network time.



The procurement clock: specification, tender, award, delivery, installation and acceptance often take 6–18 months. Start procurement in parallel with the pilot, not after it.

Public cloud: the traps nobody budgets for

Not reasons to avoid cloud — reasons to write them into the business case on day one



Egress charges

Moving microdata out of the cloud repeatedly creates a recurring, uncapped charge.



Idle resources

GPU instances left on overnight and at weekends are the top source of overspend.



Capacity & quota

The GPU type you designed for may be unavailable in your region when you need it.



Residency ≠ immunity

Data stored in-country may still be reachable by foreign legal process against the provider.



Model retirement

Vendors retire model versions on their schedule, breaking your reproducibility chain.



Convenience lock-in

Proprietary tooling is the fastest way in and the slowest way out. Prefer open formats.



Data-use terms

Get "no training on our data", log retention and sub-processor lists in the contract.



Currency risk

Bills priced in US dollars rise when the local currency weakens during the budget year.



KEY POINT Get written answers on legal access and data use before the pilot, not before renewal. A lawyer is cheaper than a confidentiality incident.

Hosting options for African statistical offices

In-country capacity is growing but concentrated — choose deliberately and ask the right questions



Own server room

Full control; limited by power, cooling and scarce platform staff.



National government data centre

Public-sector terms and shared operations; check capacity and GPU availability.



In-country commercial host

Colocation or local cloud; growing in major hubs, varies widely by country.



Hyperscaler region elsewhere

Mainly South Africa today; cross-border transfer needs a clear legal basis.

QUESTIONS TO ASK ANY HOST

- ? Where exactly are the data and backups stored?
- ? Which country's law governs access to them?
- ? What is the power autonomy and uptime record?
- ? Are GPUs available, and at what lead time?
- ? How do we get our data back if we leave?
- ? Does our statistics act permit this location?
- ? Who pays, and in which currency?



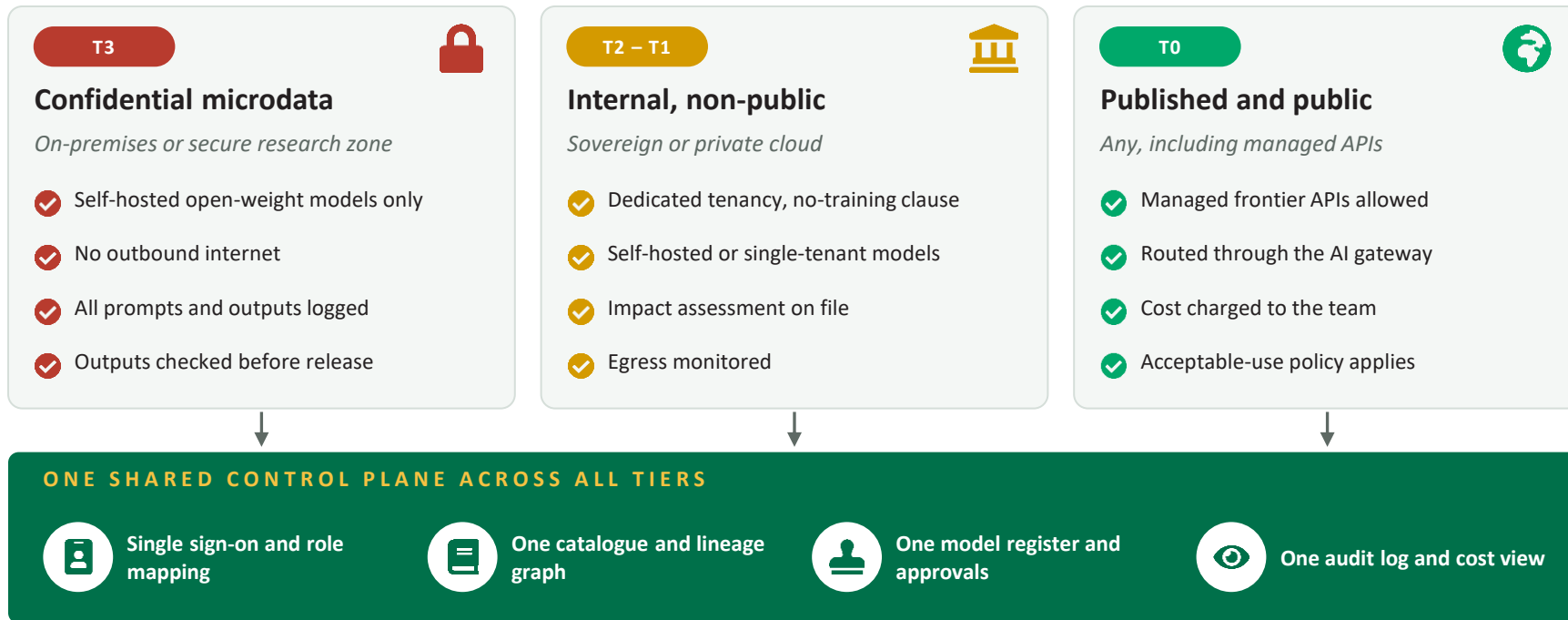
CONTEXT

46% of Africa's 211 operational data centres are in just four countries: South Africa, Kenya, Nigeria and Egypt.

Sources: Heirs Technologies study (Sept. 2025), via Ecofin Agency; ADCA, Data Centres in Africa 2026; industry reporting on Oracle's Kenya cloud region (2026).

The hybrid pattern most offices converge on

Three processing tiers, one shared control plane. The data chooses the tier, not the team



Build or buy, component by component

Rule: build what encodes your methodology; adopt what is generic plumbing

COMPONENT	DEFAULT CALL	WHY
Foundation models	ADOPT	No statistical office has a case for pre-training. Choose open-weight or proprietary.
Domain classifiers and coders	BUILD	They encode your classifications, coding rules and history. Nobody else has this data.
Data platform and catalogue	ADOPT	Mature open-source and commercial options exist. Prefer open formats.
Orchestration and scheduling	ADOPT	Airflow, Argo, Kubeflow and similar are commodities. Do not write your own.
Vector store and retrieval	ADOPT	Commodity and fast-moving; keep the embedding pipeline in your own code.
Evaluation harness	BUILD	Your gold standard, rare classes and human baseline. No vendor benchmark replaces it.
Annotation tooling	ADOPT	Buy the tool; own the guidelines, the adjudication and the resulting data.
AI gateway and confidentiality controls	BUILD / ASSEMBLE	Redaction rules and allow-lists are institution-specific. Assemble, but own the logic.



KEY POINT

Every BUILD here would still be valuable if every current AI vendor disappeared tomorrow. That is the test.

Open-weight vs. proprietary models

Not a philosophical question — a portfolio decision driven by data tier and reproducibility

	 OPEN-WEIGHT, SELF-HOSTED	 PROPRIETARY, MANAGED API
Confidentiality	Runs inside your perimeter; data never leaves	Data leaves unless a single-tenant deployment is contracted
Reproducibility	Weights can be pinned and archived indefinitely	Versions are retired on the vendor's schedule
Capability	Strong and improving; often enough for extraction and coding	Usually leads on the hardest reasoning and many languages
Cost shape	Fixed: hardware and staff, whatever the volume	Variable: per token, rising directly with use
Operations	You run serving, scaling, patching, monitoring	Vendor operates; you manage contract, quota, cost
Auditability	Full inspection of weights, prompts and path	Behavioural testing only; internals hidden
Lock-in	Low — portable formats, replaceable serving	Moderate to high, especially with vendor tooling

 **KEY POINT** Run a portfolio: open weights for confidential and reproducibility-critical work, managed APIs for exploration. Route both through one gateway so you can switch.

Exercise: where should this workload run?

Discuss each case for two minutes, then compare with the suggested answer

1

Coding 2 million census occupation answers that still contain names and addresses

Suggested answer: T3 · On-premises, self-hosted model

Identifiable microdata: classification decides before cost.

2

Translating a published statistical bulletin into French, Arabic and Kiswahili

Suggested answer: T0 · Managed API via the gateway

Public material; keep a human review for language quality.

3

A staff assistant answering questions about internal methodology manuals

Suggested answer: T1 · Private cloud under contract

Non-public but not microdata; retrieval must filter by user rights.

4

Training a crop-type classifier on public satellite imagery for two weeks

Suggested answer: Archetype C · Dedicated cloud, burst

Public inputs and a short, heavy load: rent, do not buy.

04

Governance & confidentiality

How do we protect statistical secrecy when AI opens new ways for data to leak?

IN THIS SECTION

- › New disclosure pathways
- › The legal landscape
- › Classification crosswalk
- › Technical controls
- › Model governance
- › AI in the GSBPM
- › Five failure scenarios



Seven new ways confidential data can leak

Disclosure pathways your current controls probably do not cover



Shadow AI

Staff pasting microdata into personal accounts on public chatbots. The most likely incident.



Memorisation

A model fine-tuned on microdata can repeat training records. Classify weights like their data.



Vector stores

An embedding index is a second copy of the documents, often outside original access controls.



Disclosure in outputs

A generated summary can reveal a small cell that tabular controls would suppress.



Logs and telemetry

Stored prompts and responses form a third, uncontrolled copy. Set retention on both sides.



Sub-processors

Your contract is with one vendor; the processing may run on another's infrastructure.



Inference context

Without per-user filtering, retrieval can show content a user has no right to see.



KEY POINT

Shadow AI is not solved by a memo. It is solved by a sanctioned tool good enough that the unsanctioned one is not tempting.

The legal landscape: global, African, national

Obligations differ by country, but the structure of the analysis is the same

	STATISTICS	DATA PROTECTION & AI	SECURITY & OPERATIONS
Global	UN Fundamental Principles of Official Statistics — Principle 6: individual data strictly confidential	OECD AI Principles; UNESCO Recommendation on the Ethics of AI; EU AI Act (Reg. 2024/1689) as a reference model	International security standards (e.g. ISO/IEC 27001); archiving good practice
African	African Charter on Statistics; SHaSA 2 — harmonisation and professional independence	Malabo Convention (in force June 2023); AU Data Policy Framework; AU Continental AI Strategy (July 2024)	Malabo cyber-security provisions; regional (REC) frameworks, e.g. ECOWAS data protection act
National	National statistics act: mandate, statistical secrecy, limits on where and by whom data is processed	National data protection law and authority; national AI strategy where one exists	Cyber-security law, public procurement law, archiving and retention law

Sequencing: bring legal, the data protection officer and security into the first design workshop. An objection raised at design costs an afternoon; at deployment it costs a project.

Data classification mapped to permitted infrastructure

The most useful single artefact in AI governance. Write it, have it signed, publish it

TIER	DATA	ENVIRONMENT	MODELS	REQUIRED CONTROLS
T3	Identifiable or easily re-identifiable microdata	On-premises or secure research zone only	Self-hosted open-weight from an approved list	No outbound network · full logging · output checking · named access
T2	Pseudonymised or detailed non-public data	Sovereign or dedicated-tenancy cloud	Self-hosted, or single-tenant managed under contract	Impact assessment · no-training clause · encryption with own keys · egress monitoring
T1	Internal documents, metadata and code	Private or public cloud under enterprise agreement	Managed models permitted under contract	No-training clause · single sign-on · gateway routing · retention limits
T0	Published outputs and public material	Any, including public managed APIs	Any approved model	Acceptable-use policy · gateway routing for cost and audit



WHY IT MATTERS

This table turns an endless case-by-case debate into a lookup. An analyst who knows their data is T2 already knows the environment, models and paperwork. Review it every year.

Technical controls that enforce the policy

A policy nobody can break by accident is worth more than one everybody has read



Network egress control

No route to the internet from the secure zone.



Identity and roles

Single sign-on, MFA, roles mapped to tiers.



Encryption and keys

At rest and in transit; office-held keys where needed.

HIGHEST-LEVERAGE INVESTMENT



The internal AI gateway

One audited chokepoint for every model call

- ✓ Model allow-list per tier
- ✓ Redaction and prompt checks
- ✓ Tamper-proof logging
- ✓ Quotas and budget caps
- ✓ Cost per project code
- ✓ Swap models by configuration



Access-filtered retrieval

Apply user permissions at query time, not only at indexing.



Secrets management

No passwords in notebooks or code; short-lived tokens.



Retention and deletion

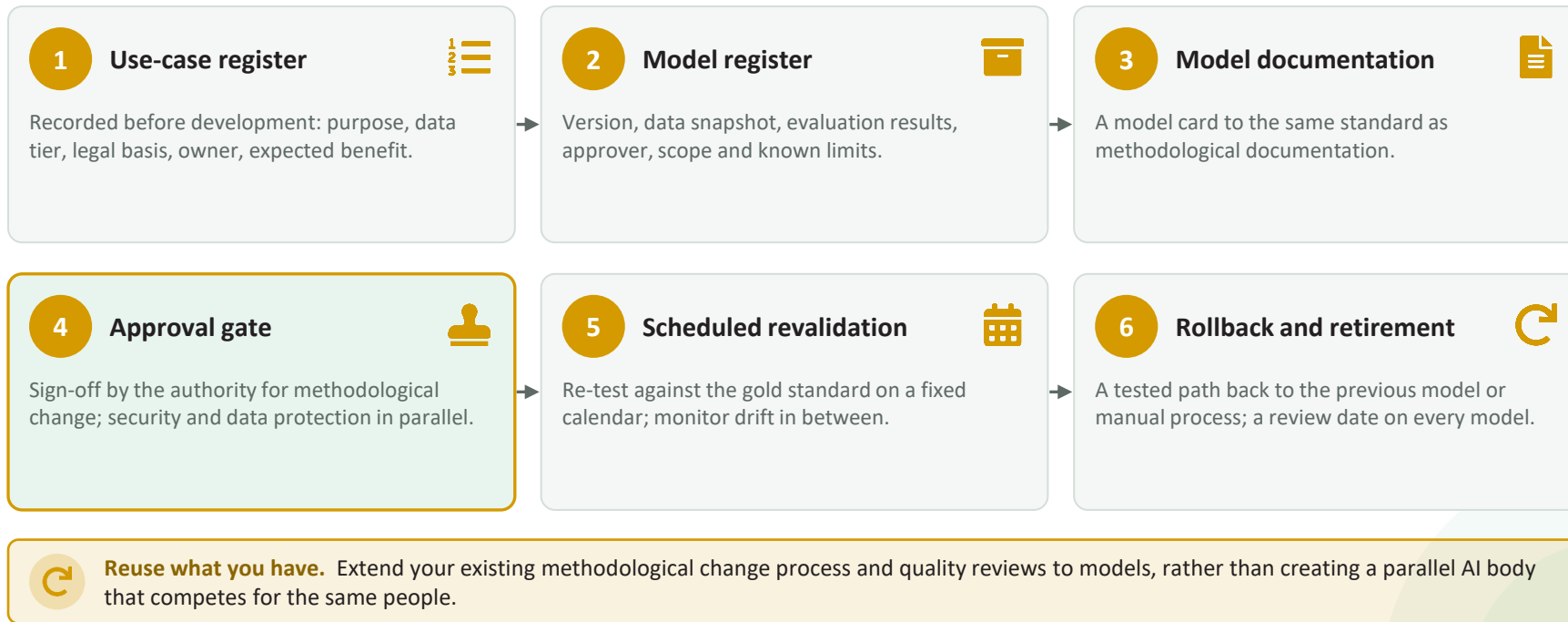
Explicit rules for prompts, logs, indexes and snapshots.



KEY POINT Build the gateway before the first production use case, not after the fifth. Retrofitting a chokepoint costs an order of magnitude more.

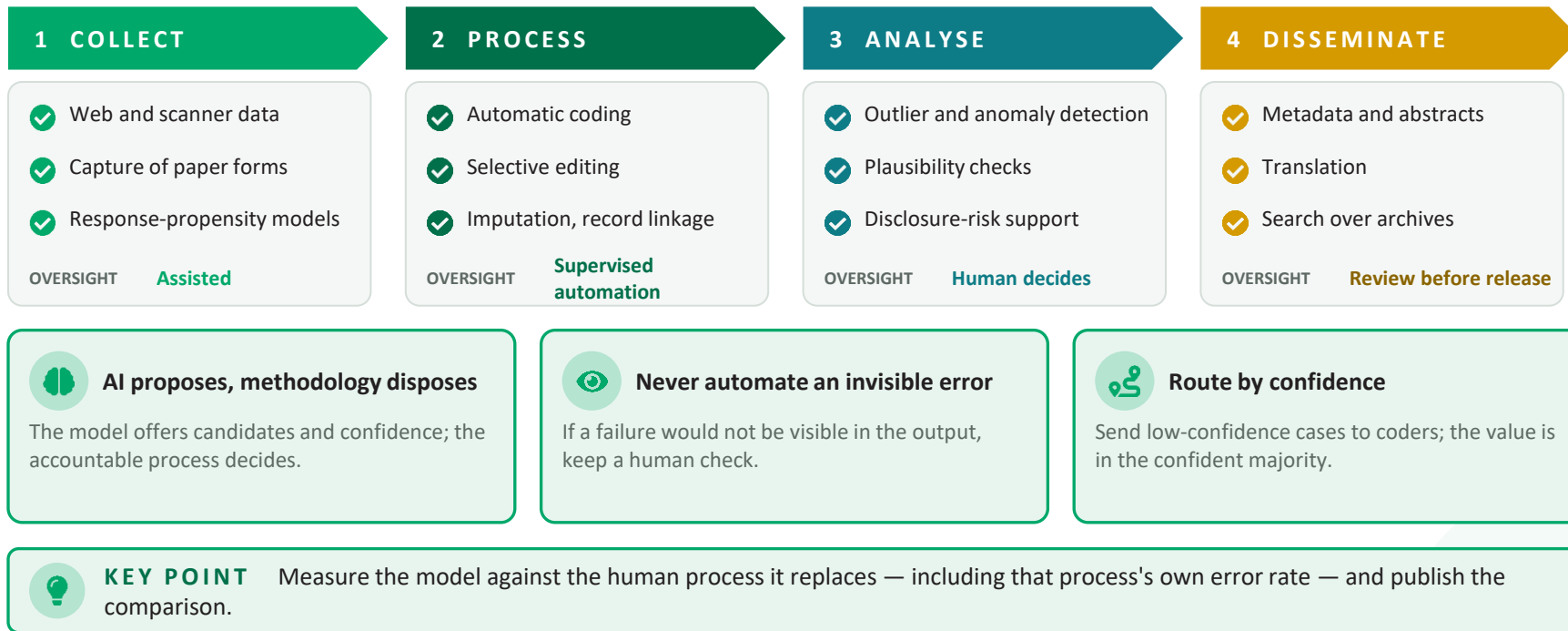
Model governance: the register and the gate

Every model behind a published statistic has an owner, a version, an approval and an expiry date



AI inside the statistical production process

Mapped onto GSBPM phases, with the level of human oversight each warrants



Five realistic failure scenarios

Each has happened somewhere. Each has a specific, affordable control

SCENARIO	CONSEQUENCE	THE CONTROL THAT PREVENTS IT
 An analyst pastes identifiable microdata into a public chatbot	Notifiable breach; loss of respondent trust; possible parliamentary interest	A sanctioned internal tool, egress controls and a short rule staff have actually read
 A vendor silently updates a model; coding shifts mid-series	The published series changes for reasons nobody can document	Pin versions; re-run the gold-standard test before any change; self-host on the critical path
 A job-title classifier quietly degrades as the labour market changes	Slow quality erosion, discovered by an external user	Drift monitoring, scheduled revalidation and results broken down by category
 An unmonitored batch job runs all weekend on a pay-per-use API	A cost overrun that damages credibility with finance	Hard budget caps at the gateway, team quotas, alerts on rate of spend
 A successful pilot cannot go live: no data platform exists	"Pilot purgatory": enthusiasm spent, nothing in production after two years	Build the platform before the second use case; require the promotion path from the start



KEY POINT

None of these controls is expensive — and all are much cheaper before the first production deployment than after the incident.

SECTION 05

05

Cost & budgeting

Where does the money really go over five years?

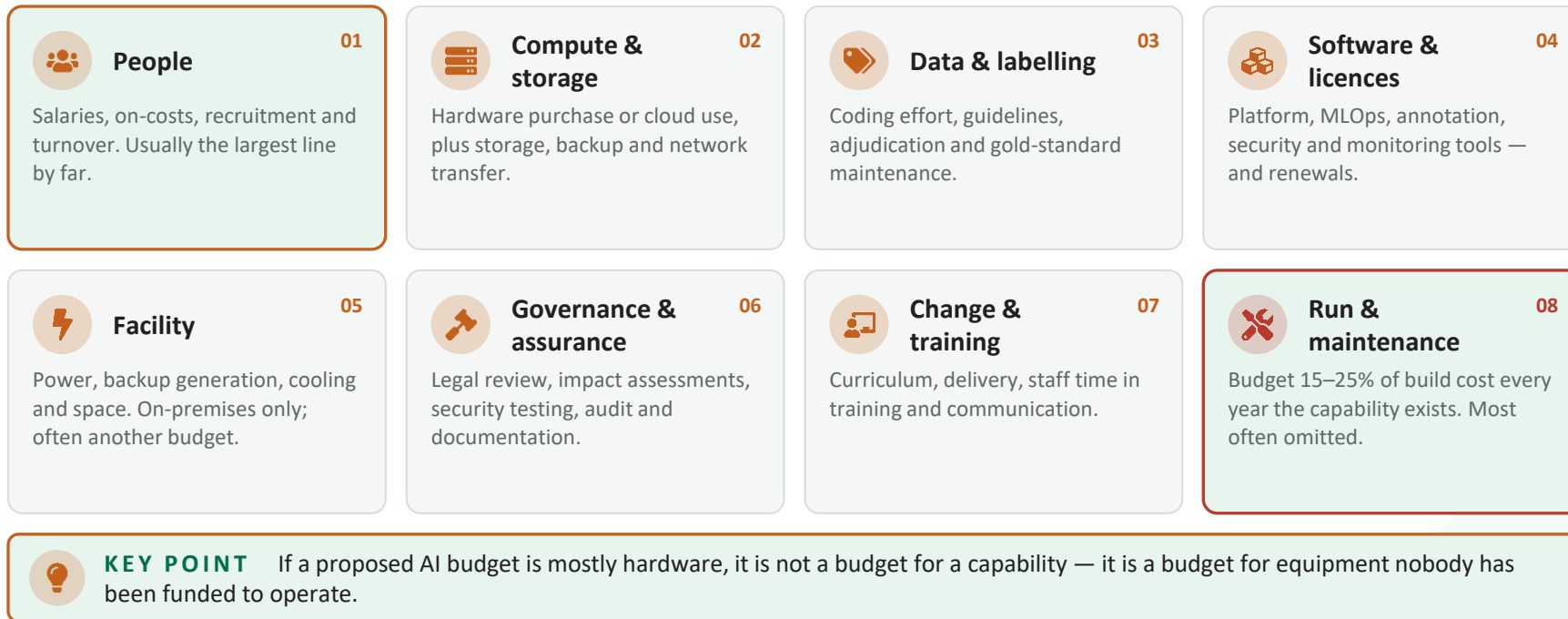
IN THIS SECTION

- › The full cost stack
- › Five-year composition
- › Three TCO scenarios
- › Break-even as volume grows
- › Where budgets break
- › Cost controls (FinOps)
- › Making the budget case



The full cost stack

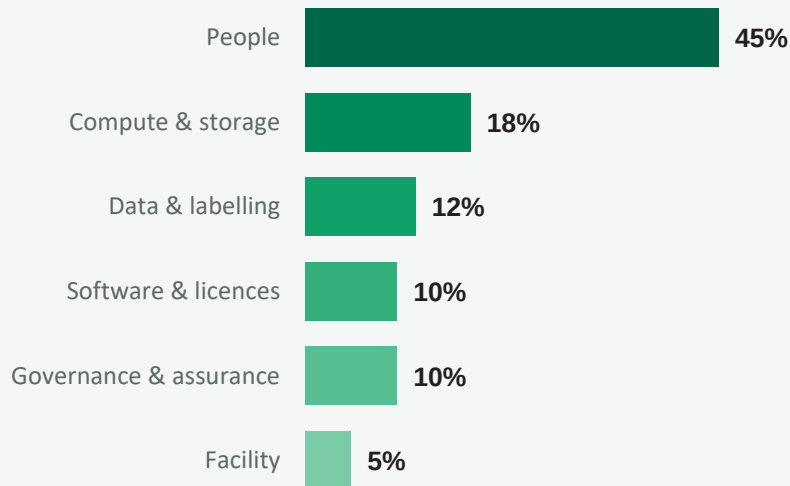
Hardware and API charges are the visible costs. They are rarely the largest



Where the five-year money goes

Illustrative composition for a mid-sized office running two or three production use cases

Share of five-year total cost (%)



People dominate

Nearly half of five-year cost is salary — so staffing moves the total more than hosting does.



Compute is a minority

Hardware plus facility rarely exceeds about a quarter of the total.



Governance is not free

Assurance is real, recurring effort. Budget it or it is taken from delivery.



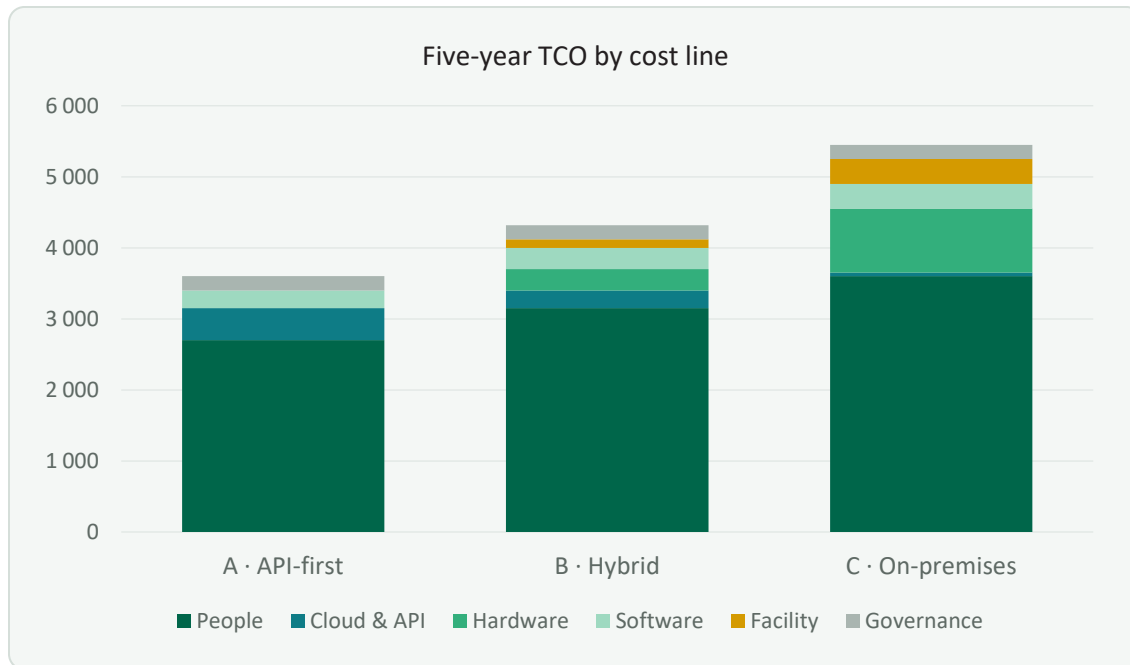
Labelling is an investment

The gold standard outlives every model trained on it.

Illustrative planning model based on typical public-sector experience — not measurements. Substitute your own unit costs before use.

Five-year cost of ownership: three scenarios

Same office, same two use cases, three deployment choices (illustrative, thousands of units)

**3,600****Scenario A**

Fastest to value; cost rises directly with usage

4,320**Scenario B**

Owned capacity for T3–T2, elastic burst for the rest

5,450**Scenario C**

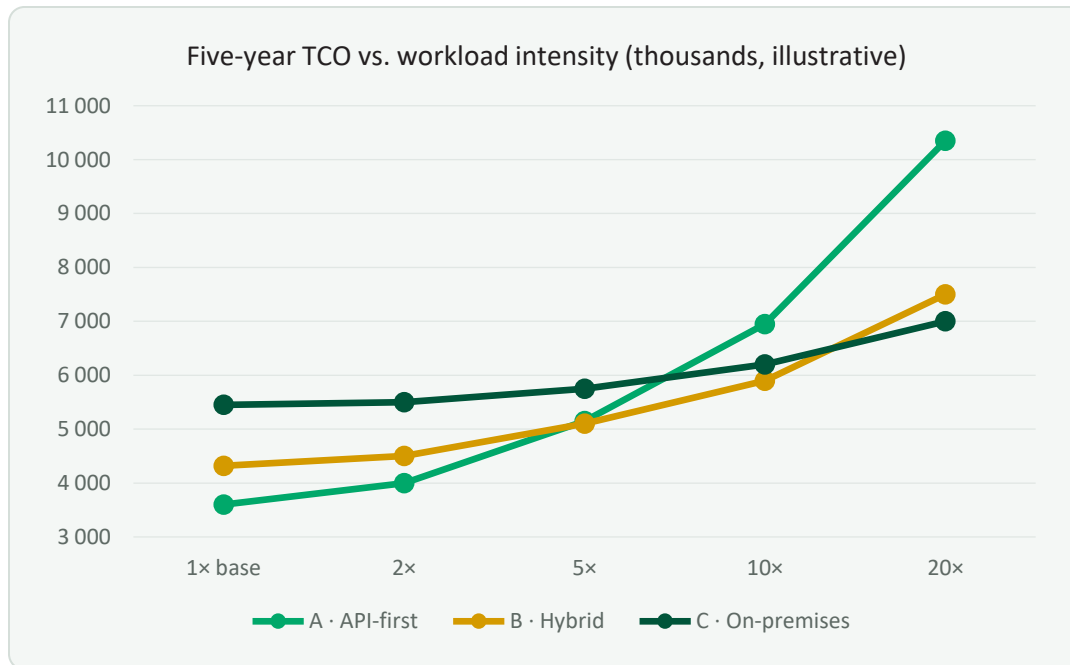
Full control; economic only at high sustained use

**Hosting moves about a third of the total.
Staffing moves more.**

Illustrative planning model: 2 use cases, 6–8 FTE, one 4-GPU node in B, a small cluster in C. Not a quotation — rebuild with local costs.

How the answer changes as volume grows

The same three scenarios at rising workload. The crossing points are the real decision points



Low volume: A wins

Pilots and modest loads are cheapest on managed services.

Mid volume: B wins

Steady load suits owned base capacity plus elastic burst.

High volume: C wins

Only sustained high use repays a full owned estate.

Classification overrides all

For T3 data, option A was never available at any volume.

Illustrative model. The shape — a flat owned-capacity line crossing a rising usage line — is robust; the crossing points depend on your costs.

Seven places the budget breaks

Each has ended, delayed or embarrassed a public-sector AI programme somewhere



Data transfer

Moving data repeatedly between tiers: uncapped and invisible until the invoice.



Idle GPUs

Development machines left on at night and weekends. Most common, easiest to fix.



Re-indexing

Each document refresh re-embeds everything; cost grows quietly with size and frequency.



Labelling underestimated

Guidelines, adjudication and re-coding after classification revisions are real effort.



"Free" assurance

Security, legal and methodological review is paid in someone's time either way.



No run budget

The project closes; the dependency remains and degrades until it fails in public.



Staff turnover

Losing the one engineer who knows the pipeline can cost months — or the capability.



KEY POINT

Write all seven into the business case as named risks with named owners. A documented risk is a managed cost.

Cost controls that work in a public office

Attribution first: you cannot manage what you cannot assign to a team and a purpose



GUARDRAILS

1

Attribute every call

Tag each request with a project code via the gateway.

2

Cap, do not just alert

Hard ceilings that stop requests — dashboards do not stop weekend overruns.

3

Automatic shutdown

Idle machines stop on a schedule unless explicitly extended.

4

Quotas and thresholds

Per-team quotas with an approval path above a limit.



EFFICIENCY

5

Smallest sufficient model

Start small or self-hosted; escalate only when tests show the need.

6

Cache and batch

Reuse results for repeated inputs; batch jobs cost less than live calls.

7

Commit once load is known

Reserved capacity saves money — only after real usage history.

8

Review cost with delivery

One monthly meeting for progress and spend, same people.

Making the budget case

A request a finance committee can approve — and an auditor can later defend



DO

What a defensible submission contains

- ✓ Frame benefits as statistical outcomes: timeliness, quality
- ✓ Show cost per record today vs. with AI, with test evidence
- ✓ Ask for a platform with recurrent funding, not a project
- ✓ Phase it: pilot → platform → scale, with a decision at each step
- ✓ Include run, refresh and decommissioning costs
- ✓ State the confidentiality position up front
- ✓ Name unknowns; propose a cost ceiling and review date



DO NOT

What gets a submission sent back

- ✗ Lead with the technology or a model architecture
- ✗ Promise headcount cuts — promise capacity redeployed
- ✗ Present accuracy without the human baseline
- ✗ Request hardware before a tested use case and labelled data
- ✗ Hide governance effort inside a delivery line
- ✗ Use vendor benchmarks as evidence about your data



KEY POINT The strongest sentence in any AI business case is a measured comparison with the current process, on your own data. Everything else is assertion.

SECTION 06

06

People & roadmap

Who do we need — and in what order should we build?

IN THIS SECTION

- › The team you need
- › Staffing reality
- › What each group must learn
- › Five-level maturity model
- › A 24-month roadmap
- › Anti-patterns
- › Readiness checklist
- › Ten takeaways



The team you actually need

Indicative for a mid-sized office with two or three production use cases. Scarcity, not headcount, is the message

ROLE	FTE	WHAT THEY OWN	MARKET
 Use-case owner	0.5–1	The statistical problem, success criteria, go / stop decision	Existing staff
 Methodologist with ML skills	1–2	Evaluation design, human baseline, defensibility of outputs	Grow internally
 Data engineer	1–2	Pipelines, data platform, snapshots, lineage, data quality	Scarce — biggest gap
 ML engineer / data scientist	1–2	Modelling, fine-tuning, retrieval design, evaluation harness	Competitive
 Platform / MLOps engineer	1–2	Serving, gateway, environments, monitoring, GPU operations	Scarce
 Security & data protection	0.3	Classification decisions, impact assessments, controls, incidents	Existing staff
 Subject-matter annotators	Varies	Labelling, guidelines, adjudication, gold-standard upkeep	Your hidden advantage
 Change & training lead	0.5	Curriculum, communication, adoption, effect on existing roles	Existing staff

 **KEY POINT** Most offices over-recruit data scientists and under-recruit data and platform engineers. The result: excellent notebooks that never reach production.

Staffing reality in a public statistical office

You will not win on salary. You can win on almost everything else



What you cannot offer

Private-sector salaries for experienced ML and platform engineers. Accept it and design around it.



What you genuinely can offer

A public mission, uniquely rich data, rigour, publication, training budgets, stability and predictable hours.



The dependency you cannot afford: fully outsourcing the platform leaves the office unable to evaluate, run or exit its own systems. Buy tools and help — never the understanding of how your statistics are made.

FIVE STRATEGIES THAT WORK



Grow from within

Turn methodologists into applied ML practitioners — they already know the data.



Partner rather than purchase

Universities, training centres, peer offices and development partners.



Share across government

One national platform or gateway team is easier to staff than one per agency.



Contract with transfer duties

Make documented knowledge transfer a deliverable, checked before final payment.



Build a technical career path

Senior technical grades, so engineers need not move into management to progress.

What each group has to learn

Three different curricula. The failure mode is one generic introduction for everybody



Statisticians & methodologists

- ✓ How supervised learning differs from model-based inference
- ✓ Evaluation: precision, recall, calibration, thresholds
- ✓ Python or R, version control, reproducible environments
- ✓ Enough about retrieval and prompting to judge reliability
- ✓ Bias, drift and how models degrade quietly



IT, platform & security

- ✓ Containers and orchestration (Kubernetes or Slurm)
- ✓ GPU operations: drivers, scheduling, sharing, monitoring
- ✓ Model serving, gateways, vector stores
- ✓ CI/CD for models, data and prompts
- ✓ AI security: prompt injection, leakage, supply chain



Management & oversight

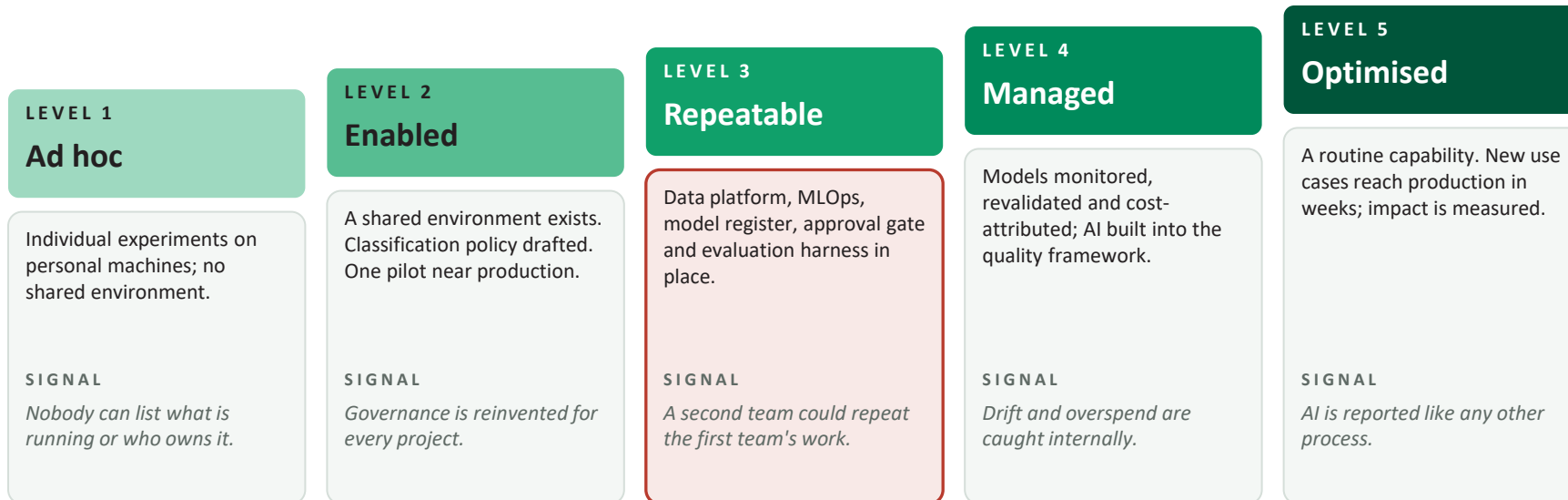
- ✓ What AI can and cannot do, without vendor framing
- ✓ How to read an evaluation report
- ✓ Which cost drivers actually move the total
- ✓ Legal duties: confidentiality, data protection, transparency
- ✓ How and when to stop a project



Format matters more than content. Generic online courses show low completion and little transfer. What works: learning tied to a real internal use case, with protected time, a mentor and a useful deliverable.

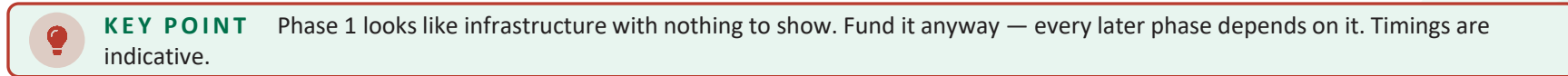
A five-level maturity model

Be honest about the starting point. Most statistical offices sit between levels 1 and 2



KEY POINT Assess against your weakest element, not your strongest. The jump from level 2 to 3 is the expensive one — it is where a pilot culture becomes an operating discipline.

Do not begin a phase until the previous gate is genuinely met



Eight anti-patterns

Recognisable ways a well-intentioned AI programme in a statistical office fails



Hardware before use case

GPUs bought on a strategy document, idle for 18 months while the data platform is built.



Hardest problem first

Confidential microdata for a flagship statistic as the first attempt: every constraint at once.



Strategy, no platform

A well-received document and a steering group — but nowhere to build anything.



The single hero

One person understands the whole pipeline. A resignation letter becomes an outage.



Pilot purgatory

Proof of concept after proof of concept, applauded but never promoted to production.



Governance afterwards

Controls designed once the system is live can only be advisory — and cost more.



Output treated as truth

Generated answers enter a statistical product with no deterministic or human check.



Measured against perfection

Rejecting a system better than the manual process, whose error rate was never measured.



KEY POINT

Seven of these eight are sequencing failures, not technology failures. The order of the work matters more than the choice of tools.

Readiness checklist

Ten yes/no questions. Fewer than seven "yes" answers means you are not ready to scale beyond a pilot

1

A classification crosswalk is approved, and staff can find it

6

Every production model has an owner and a review date

2

A governed data platform with catalogue and lineage exists

7

Costs are attributed, and hard caps are enforced

3

A frozen gold-standard test set with a measured human baseline exists

8

Legal, DPO and security were involved at design stage

4

Every model call passes through one audited gateway

9

Every production use has a tested fallback

5

DEV, TEST and PROD are separated, with named approvers

10

Run and refresh costs are funded beyond the current project

SCALE WHEN

 ≥ 7

out of 10 answers are "yes"

Nine of the ten cost time and attention, not money. None needs an extra GPU.

Ten things to take away

If you remember nothing else from this briefing

01

AI in a statistical office is a platform with legal obligations, not a project with a model in it.

02

The model is a small fraction of the system. Everything around it is the work — and the cost.

03

The binding constraint is labelled data and a governed data platform, not GPUs.

04

Data classification decides where workloads run. Cost only chooses among permitted options.

05

Write the classification crosswalk, have it signed and publish it.

06

Build one audited AI gateway before the first production use case.

07

Reproducibility is mandatory: pin generative models or keep them off the critical path.

08

People are about half the five-year cost. Staffing moves the budget more than hosting.

09

Measure against the current human process, including its error rate, and publish it.

10

Sequence deliberately: foundations, platform, one use case, then scale.

Sources and further reading

Standards, frameworks and data referred to in this briefing



Official statistics frameworks

- UN Fundamental Principles of Official Statistics (UNGA res. 68/261, 2014) — Principle 6
- UNECE HLG-MOS: GSBPM, GAMSO, GSIM and machine-learning project outputs
- African Charter on Statistics (AU, 2009)
- Strategy for the Harmonisation of Statistics in Africa (SHaSA 2)
- European Statistics Code of Practice
- Statistics Canada, Framework for Responsible Machine Learning Processes



Law, policy and technical

- AU Convention on Cyber Security and Personal Data Protection (Malabo, 2014; in force 8 June 2023)
- AU Data Policy Framework; AU Continental AI Strategy (endorsed July 2024)
- Regulation (EU) 2024/1689 (EU AI Act)
- OECD AI Principles; UNESCO Recommendation on the Ethics of AI (2021)
- D. Sculley et al., "Hidden Technical Debt in Machine Learning Systems", NeurIPS 2015



African context data

- IEA, Financing Electricity Access in Africa (2025)
- Africa Data Centres Association, Data Centres in Africa 2026: The Economic Report
- BCG, Advancing Africa's AI and Digital Economy (2026)
- Heirs Technologies, Africa's Digital Leap (Sept. 2025), reported by Ecofin Agency
- World Bank Group & AfDB, Mission 300 (2024)
- AU Executive Council, 45th Ordinary Session, Accra (July 2024)